Surrogate Neural Network Guided Evolutionary Optimisation for the P-Centre Problem

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DRDO Research Internship Report

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Abstract

This work presents a hybrid metaheuristic optimisation framework integrating an evolutionary strategy with a neural network surrogate model for solving the P-Centre Problem. The algorithm employs rank-based probabilistic selection to preserve population diversity and mitigate premature convergence. A self-adaptive Gaussian mutation mechanism enables individual candidates to dynamically adjust their mutation strength, enhancing search efficiency. Training data collected during evolutionary runs—consisting of normalised candidate features and associated fitness values—are used to train a deep neural network surrogate that approximates the computationally expensive fitness evaluation function. This surrogate enables rapid fitness prediction during subsequent optimisation runs, significantly reducing computational cost. Experimental evaluation shows that the hybrid approach achieves solution quality comparable or superior to baseline methods with substantial runtime reductions. These results demonstrate the efficacy of coupling adaptive evolutionary search with surrogate modelling for scalable and robust P-Centre Problem optimisation.

1. Introduction

Combinatorial optimisation problems are central to operations research and have important applications in logistics, network design, and resource allocation. Among these, the *P-Centre Problem*—which involves selecting a subset of facilities to minimise the maximum distance any client must travel—presents significant computational challenges due to its combinatorial complexity.

Metaheuristic algorithms have been widely adopted to provide efficient approximate solutions for such problems, thanks to their ability to explore large solution spaces without exhaustive search. However, these methods often require extensive fitness evaluations, which can be computationally expensive, particularly when the fitness function itself is complex.

In this study, we propose a hybrid optimisation framework that combines an evolutionary strategy with a neural network surrogate model to address these challenges. The evolutionary strategy includes rank-based probabilistic selection to maintain diversity within the population and employs self-adaptive Gaussian mutation to dynamically adjust the search step size for each individual candidate. The neural network surrogate is trained on historical fitness data obtained during optimisation runs, enabling fast approximation of fitness evaluations in subsequent iterations.

This approach aims to balance the trade-off between computational efficiency and solution quality, making it scalable for large problem instances. The proposed method is evaluated on instances of the P-Centre Problem, demonstrating its capability to find high-quality solutions efficiently.

2. Literature Review

The P-Centre Problem, a classical combinatorial optimisation challenge, has received significant attention in operations research due to its applicability in facility location and network design. Multiple variants and formulations of the problem have been studied extensively in terms of theoretical complexity, exact approaches, and heuristic solutions.

Early exact methods primarily relied on integer and mixed-integer linear programming formulations and branch-and-bound algorithms [7, 8]. While exact methods guarantee optimality, their applicability is limited to small and medium-sized instances due to computational complexity.

To address scalability and computational limitations, a rich body of work has focused on metaheuristic algorithms. These approaches—including genetic algorithms, tabu search, variable neighbourhood search, and memetic algorithms—provide approximate solutions by iteratively exploring the solution space using stochastic and adaptive mechanisms [9, 10]. Metaheuristics are particularly effective for large-scale problems but rely heavily on the number of fitness evaluations, which often dominate overall computational cost.

Increasingly, surrogate modelling techniques have been combined with metaheuristic optimisation to alleviate expensive evaluations. Neural networks, due to their ability to learn complex nonlinear mappings, have been widely utilised as surrogate models to approximate fitness functions [11, 12]. Surrogate-assisted evolutionary algorithms leverage these models to guide the search process efficiently by reducing the frequency of costly direct evaluations. Various strategies for integrating surrogates—including periodically retraining models with evolving data—have demonstrated improved convergence speed without significant degradation in solution accuracy [13, 14].

Recent works have explored self-adaptive mutation mechanisms within evolutionary strategies to dynamically balance exploration and exploitation during the optimisation process [15]. Similarly, rank-based selection schemes have been proposed to maintain population diversity and prevent premature convergence by probabilistically selecting candidate solutions relative to their rank rather than raw fitness values [16].

This report builds upon these strands of research by proposing a hybrid metaheuristic framework that integrates rank-based selection and self-adaptive Gaussian mutation within an evolutionary strategy, augmented by a neural network surrogate model. This combination aims to efficiently solve large-scale P-Centre Problem instances, reducing computational overhead while maintaining high solution quality. Comparative evaluation elucidates the advantages of this integrated approach over conventional methods.

3. Proposed Method

3.1 Problem Setup and Inputs

We address the discrete P-Centre Problem, where the goal is to select p centres from n candidate nodes so as to minimise the maximum distance any node has to its closest centre. The problem input consists of:

- **Number of nodes** (n): Total locations in the problem space.

- **Number of centres** (p): Number of centres to be selected.
- **Distance matrix** ($\mathbf{D}$): An $n \times n$ matrix where each element d_{ij} is the distance between node i and node j.

The fitness (quality) of any candidate solution S (a set of p centres) is measured by:

$$f(S) = \max_{i=1}^{n} \min_{c \in S} d_{i,c} \tag{1}$$

The problem can be formally stated as the integer programme:

$$\begin{aligned} & \min \; Z \\ \text{subject to} \quad & \sum_{j=1}^{n} x_{ij} = 1, && \forall\, i = 1, \ldots, n \\ & \sum_{j=1}^{n} y_j = p \\ & x_{ij} \leq y_j, && \forall\, i, j \\ & \sum_{j=1}^{n} d_{ij}\, x_{ij} \leq Z, && \forall\, i \\ & x_{ij} \in \{0,1\}, \quad y_j \in \{0,1\} \end{aligned} \tag{2}$$

where $y_j = 1$ if a centre is placed at node j (and 0 otherwise), $x_{ij} = 1$ if node i is assigned to centre j (and 0 otherwise), and Z is the minimised covering radius.

3.2 Distance Matrix Generation

The distance matrix $\mathbf{D}$ is generated by computing pairwise Euclidean distances between nodes, or by loading a prespecified problem-specific distance structure. This matrix enables rapid fitness evaluation of candidate centre sets by direct lookup, avoiding repeated distance computations during optimisation.

3.3 Algorithm Overview

The proposed algorithm combines an evolutionary search strategy with a neural network surrogate model to optimise both computational efficiency and solution quality. The high-level flow is:

1. **Initialisation** — define an initial population of random candidate solutions and their associated mutation strengths; load the distance matrix.
2. **Evolutionary search** — for T generations, apply rank-based parent selection, self-adaptive Gaussian mutation, fitness evaluation, and rank-based survivor selection.
3. **Surrogate training** — after T generations, train a deep neural network $\mathcal{N}$ on the accumulated dataset of (feature, fitness) pairs.
4. **Surrogate-guided optimisation** — in subsequent runs, replace the expensive direct fitness function with rapid surrogate predictions $\hat{f}(S) = \mathcal{N}(\phi(S))$.

3.4 Evolutionary Algorithm Details

3.4.1 Population Initialisation

The initial population $P^{(0)} = \{S_1^{(0)}, S_2^{(0)}, \ldots, S_{n_i}^{(0)}\}$ consists of n_i candidate solutions. Each $S_i^{(0)} \subseteq \{1, \ldots, n\}$ is generated by selecting p unique centres uniformly at random:

$$S_i^{(0)} = \text{sorted}\Big(\text{random_sample}(\{1, \ldots, n\},\, p)\Big).$$

This ensures diverse exploration at the start of optimisation.

3.4.2 Fitness Evaluation

Given the distance matrix $\mathbf{D} = [d_{ij}]$, the fitness of candidate S_i is:

$$f(S_i) = \max_{1 \le k \le n} \min_{c \in S_i} d_{k,c}.$$

Lower fitness corresponds to better coverage of the nodes.

3.4.3 Rank-Based Selection

Candidates are ranked in ascending order of fitness. The probability p_i of selecting candidate S_i as a parent is proportional to its rank r_i:

$$p_i = \frac{r_i}{\sum_{j=1}^{n_i} r_j},$$

where the best candidate receives rank $r = n_i$. This probabilistic scheme promotes diversity and guards against premature convergence.

3.4.4 Self-Adaptive Gaussian Mutation

Each selected parent $S_i^{(t)}$ produces offspring through Gaussian mutation with a self-adaptive mutation strength. The mutation strength evolves as:

$$\sigma_i^{(t+1)} = \sigma_i^{(t)} \cdot \exp\Big(\tau \cdot \mathcal{N}(0,1)\Big), \tag{3}$$

where τ is a learning rate parameter and $\mathcal{N}(0,1)$ is a standard normal random variable. Each centre $c \in S_i$ is then perturbed as:

$$c' = \text{clip}\Big(\text{round}\Big(c + \mathcal{N}(0,\, \sigma_i^{(t+1)})\Big),\, 1,\, n\Big),$$

ensuring valid node indices and uniqueness of centres. This mechanism dynamically tunes search step sizes, balancing local exploitation and global exploration.

3.4.5 Survivor Selection

Parents and offspring are combined into a temporary pool. Rank-based selection is applied to choose the $n_{\max}$ fittest solutions, which—together with their updated mutation strengths—form the next generation $P^{(t+1)}$.

3.5 Neural Network Surrogate

During evolutionary optimisation a dataset

$$\mathcal{D} = \Big\{(\phi(S_i),\ f(S_i))\Big\}$$

is collected, where $\phi(S_i)$ is a feature vector representing candidate S_i and $f(S_i)$ is its true fitness. The surrogate model $\mathcal{N}$ is trained to approximate the fitness function:

$$\hat{f}(S) = \mathcal{N}(\phi(S);\, \theta),$$

where θ denotes network parameters optimised by minimising the mean squared error (MSE):

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(\phi(S_i),\, f(S_i)) \in \mathcal{D}} \Big(f(S_i) - \hat{f}(S_i)\Big)^2. \tag{4}$$

The model is periodically retrained with the most recent data to improve its approximation of the evolving fitness landscape. Once trained, surrogate predictions replace direct evaluations:

$$f(S) \approx \hat{f}(S),$$

dramatically reducing runtime without significantly compromising solution quality.

3.6 Iteration and Termination

The algorithm iterates for a fixed number of generations T. After T iterations the neural network surrogate is fully trained on the accumulated dataset, and the surrogate-guided optimisation phase begins. The process terminates when T generations are exhausted or a convergence criterion is met (e.g. no improvement over successive generations). This design ensures a practical balance between solution-space exploration and computational efficiency.

4. Simulation Results

4.1 Background on Invasive Weed Optimisation (IWO)

Invasive Weed Optimisation (IWO) is a nature-inspired metaheuristic modelled after the colonising behaviour of weeds [1]. It simulates the way weeds adapt, reproduce, and compete in an environment. Better solutions produce more seeds, which spread around their parent locations according to a Gaussian distribution whose standard deviation decreases over iterations—enabling the algorithm to balance exploration and exploitation. The population is selectively reduced each iteration to maintain only the best solutions, mimicking natural selection.

4.2 Experimental Results

Table 1 summarises experiments across varying problem sizes (N) and centre counts (P). The proposed NN-surrogate approach is compared against baseline IWO in terms of runtime (seconds), solution fitness (covering radius), and surrogate model quality (MSE, MAE, R^2).

Table 1: Comparison of IWO and NN-Surrogate across problem instances.

2*N	2*P	**IWO (Baseline)**		**NN Surrogate**		Training 2*Time (s)	2*MSE	2*MAE	2*R^2
		Time (s)	Fitness	Time (s)	Fitness				
10	3	0.05	129	1.51	141	10.84	260.77	8.71	0.117
100	3	0.33	159	1.61	170	11.25	254.75	10.40	0.459
100	10	0.60	90	1.63	111	9.38	73.37	4.57	0.616
500	3	1.97	191	1.44	234	11.89	104.40	5.41	0.470
500	10	4.20	120	1.61	144	10.31	36.55	3.05	0.513
500	100	22.25	40	3.50	48	9.39	50.23	3.20	0.603
1000	3	3.86	203	0.60	216	15.75	150.62	9.09	0.668
1000	10	12.28	129	0.72	153	11.33	54.08	4.52	0.603
1000	100	67.94	43	3.54	50	9.22	11.89	1.94	0.486
3000	3	12.05	216	1.67	233	14.29	51.38	4.51	0.649
3000	10	38.99	139	1.73	157	15.16	53.96	4.78	0.587
3000	100	371.03	49	3.11	63	8.74	14.54	2.38	0.576

4.3 Discussion

The experimental results demonstrate that integrating a neural network surrogate model within the IWO framework effectively accelerates the optimisation process while maintaining competitive solution quality. Across varying problem sizes and centre counts, the surrogate model consistently reduces runtime compared to pure IWO—sometimes by an order of magnitude—illustrating its computational advantage.

Although the surrogate's predicted fitness occasionally deviates from the true fitness (as evidenced by the MSE, MAE, and R^2 metrics), it successfully guides the search towards high-quality solutions. Predictive accuracy improves with larger training datasets, as reflected in decreasing MSE and MAE and increasing R^2 values for larger problem instances. Notably, for moderate scales (e.g. $N = 100$, $P = 10$), the surrogate delivers both rapid evaluation and reasonable fitness approximation, enabling substantial speedups without significant loss in solution quality.

These findings highlight the practical trade-off between surrogate model accuracy and computational efficiency, underscoring the surrogate's role as a powerful heuristic accelerator in large-scale combinatorial optimisation.

5. Conclusion

The developed neural network surrogate model substantially enhances the efficiency of the optimisation process for the large-scale P-Centre Problem by providing rapid fitness evaluations. By approximating the computationally intensive objective function, the surrogate allows the search algorithm to explore the solution space more quickly while maintaining solution quality.

The observed predictive accuracy—supported by validation metrics such as MSE, MAE, and

R^2—indicates that although surrogate fitness predictions are approximate, they successfully guide the search toward promising solutions. Training the surrogate with diverse and sufficiently large datasets further improves its predictive capabilities, demonstrating scalability across problem sizes. Overall, this surrogate-assisted optimisation framework presents a practical and effective strategy for accelerating combinatorial optimisation tasks without significant compromise on solution accuracy.

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